

The Operator's Pocket Inspection

Capturing, Scanning, and Maintaining

Your Property with Smartphone,

LiDAR, Photogrammetry, and AI

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CHAPTER 01

Why this matters now

Five years ago this article would have been aspirational. Today it's an afternoon project for a property owner with an iPhone Pro and a credit card. The reason is convergence — small LiDAR sensors shipped in 2020, photogrammetry apps went consumer-grade around 2022, and vision-language AI models in 2023–2025 suddenly became good enough to read a property photo and tell you what's wrong with it. Each piece was useful on its own. Together they let one operator do the work of three inspection visits.

The traditional inspection pattern has stayed the same for fifty years. An operator walks the property every quarter, sometimes more. They write observations in a notebook. They flag a handful of items. They hand the notes to the operations lead, who dispatches vendors the following week. The next walk runs into the same items because nobody wrote them up systematically, and the institutional memory of the property lives in two heads — the operator's head and the lead's head. When one of them leaves, the knowledge leaves with them.

The modern stack inverts the pattern. You walk the property once with a phone and a photogrammetry app — that walk becomes a 3D model, a set of high-resolution photos, and a LiDAR scan. You run the photos through a vision-language AI and you get a written analysis, ordered by severity, with recommended actions and due dates. You push that analysis into a checklist. Six months later, you walk the property again the same way and the AI tells you what's changed. The 3D model is now the property's institutional memory — it survives personnel changes, vendor transitions, and lease disputes. The AI does not replace your judgment. It gives your judgment a body of evidence to work from.

What you can do today with a phone, a free app, and an AI subscription would have cost \$15,000 in vendor fees three years ago. That's not a budget line to underinvest in.

CHAPTER 02

What your smartphone already does

The hardware requirement is shorter than you think. If your phone is from 2020 or later, you already have roughly seventy percent of what's needed to run this workflow. The remaining thirty percent is software.

Device	Depth capability	Recommended app	Cost
iPhone 12 Pro and later (13 Pro, 14 Pro, 15 Pro, 16 Pro)	LiDAR sensor, ~5m range indoors, ~3m reliably outdoors at distance. Apple ARKit framework.	Polycam, Scaniverse	Free / freemium
iPad Pro 2020+	Same LiDAR. Useful for tripod-mounted interior captures.	Polycam, Scaniverse, KIRI Engine	Free / freemium
Google Pixel 6 Pro and later (some models)	Time-of-Flight depth sensor. ARCore depth API.	Polycam (Android), Scaniverse (Android)	Free / freemium
Samsung Galaxy S20+ and later (selected models)	ToF on some models. Otherwise software-derived depth.	Polycam (Android), Scaniverse (Android)	Free / freemium
Any 2020+ smartphone	No direct depth sensor, but high-resolution stills, 4K video and GPS tagging.	Photogrammetry apps (software-derived depth)	Free

The LiDAR sensor on the iPhone Pro is the single most underrated feature in this stack. It is not marketing — the depth data it produces is direct, not estimated. That makes photogrammetry processing faster and the resulting models substantially more accurate. If you are buying a phone tomorrow for this workflow, that is the spec to optimise for.

If you carry an older phone without LiDAR, you can still run the workflow with software-only photogrammetry. Quality is lower. Capture time runs longer. Acceptable for occasional inspections, frustrating for full-property scans. Treat the phone upgrade, if you make one, as a \$700 to \$1,300 capital purchase that pays for itself in year one.

If your phone is from 2020 or later, you already have ~70% of what this workflow needs. The remaining 30% is software and about \$20/month in subscriptions.

CHAPTER 03

The capture stack: hardware and apps

Capture stack means the set of tools you walk the property with. Every tool below is something an operator can buy, install and run today. Naming the winners here matters — there are dozens of photogrammetry apps on the market and most of them are toys.

Photogrammetry and LiDAR apps

- **Polycam** (iOS, Android; freemium). Photogrammetry plus LiDAR fusion. Best balance of quality and ease of use for property owners. Free tier works for a single shopping center. Pro tier at approximately **\$15/month** unlocks unlimited captures, higher export resolutions and commercial-use licensing. This is the default recommendation for an operator building a new workflow.
- **Scaniverse** (iOS, Android; free). Photogrammetry-focused, recently added Gaussian splatting. Excellent visual quality on the model output. The free tier is sufficient for personal use; commercial use has a separate licensing path.
- **KIRI Engine** (iOS, Android; freemium). Mobile photogrammetry with a "scan walkthrough" mode that's particularly good for capturing entire interiors in a single session.
- **RealityCapture** (Windows; free for non-commercial use, approximately **\$3,500** for a perpetual commercial licence). Desktop-grade photogrammetry from Capturing Reality. Operator-grade accuracy. Use this when Polycam's results are not accurate enough for the survey work — typically you only need this if you are producing as-built documentation for a lender.
- **Metashape** (Windows, macOS; approximately **\$179** standard, **\$3,499** professional). Pro-grade photogrammetry, used by surveyors. Overkill for most operators but worth knowing about because it is the industry standard for large commercial surveys.

Drones for exterior, roof and large sites

- **DJI Mavic 3 Enterprise with RTK module**. Approximately **\$5,500**. Survey-grade accuracy on positional data. Use this when you need a defensible aerial map for a lender, insurer or board member.
- **Skydio 2+**. Better for autonomous close-quarter capture. Acts as a flying tripod that follows the building envelope. Strong on tight site layouts where a Mavic cannot easily navigate.
- **DJI Mini 4 Pro**. Approximately **\$759**. Adequate for visual photogrammetry — not survey-grade, but the model output is good enough for a maintenance walkthrough and visual recordkeeping. The default drone for an operator who is not doing survey work.

Hardware beyond the phone

- **Ricoh Theta Z1** 360° camera. Approximately **\$1,000**. Best in class for fast interior panoramas. Output is a single 360° image you can navigate in any browser. Excellent for "this is what the common-area bathroom looked like on 14 August 2026" archival snapshots.

- **Insta360 X4.** Approximately \$400. Cheaper alternative to the Theta Z1. Less resolution but a much better price point for the same use case.
- **Matterport Pro3.** Approximately \$5,500. Dedicated LiDAR scanner. The industry-leading tool for a fully automated walkthrough. Buy only when the phone-plus-Polycam workflow stops producing model quality that meets your lender's expectations — usually year two or three of operating.

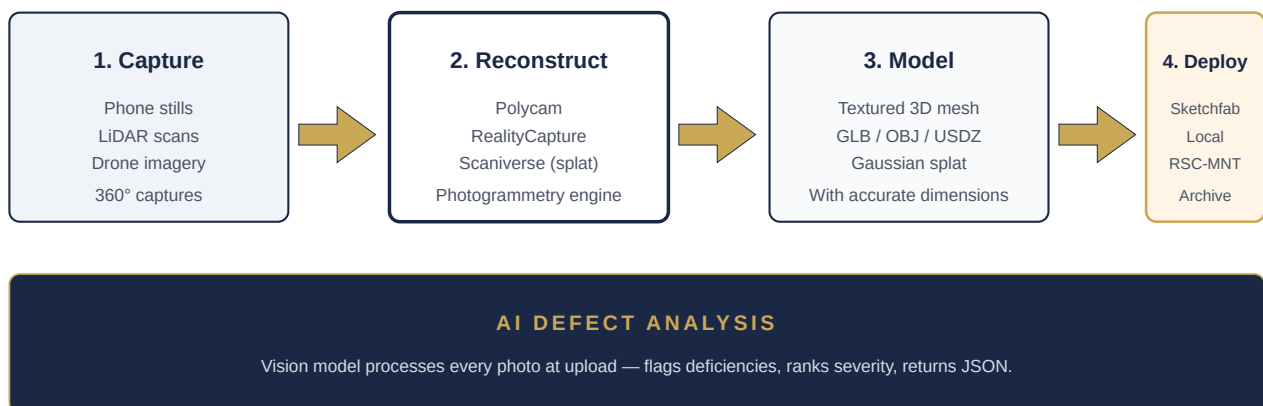
Buy in the order of use, not feature completeness. Phone and Polycam first. Drone second. Scanner last — only when the phone capture stops being good enough.

CHAPTER 04

The processing pipeline

Once you have captured, the next step is reconstruction — turning raw photos and depth data into a 3D model you can revisit and share. The pipeline below is what an operator with a phone and a credit card can run in 2026 without a surveyor's license or a \$50,000 software licence.

- **Reconstruction.** Polycam and RealityCapture generate a textured 3D model from your photos and LiDAR scans. Output formats: `GLB` (web-native), `OBJ`, `FBX` (animation), and `USDZ` (Apple AR). Polycam's LiDAR fusion mode is what gives you accurate dimensions on textured models — useful when a lender asks "how big is that vacant unit?" in the middle of a call.
- **Hosting.** Sketchfab hosts models in any of the formats above. Free tier exposes them publicly; the paid tier from approximately `$15/month` lets you keep the models private while still sharing a link. Embed anywhere — your owner report, your lender memo, your board deck.
- **Gaussian splatting.** This is the 2024 to 2025 step-change in mobile photogrammetry. Luma AI's web tools now support Gaussian splatting natively — a format that produces photorealistic walkthroughs without a gaming-grade GPU. Input is a set of photos; output is a navigable scene that looks like a video but is actually rendered from the photos themselves.



The pipeline above is the modern operator's flow. Capture once on site, run the model at your desk, push the results into your RSC-MNT (Maintenance Control Pack) bucket. Repeat quarterly. The model becomes the property's institutional record.

The output of this stage is a model you can re-visit for the next five years. Every inspection you do later with the same tool becomes a delta against this baseline. That is the structural advantage — you stop rebuilding memory every time you walk the property.

CHAPTER 05

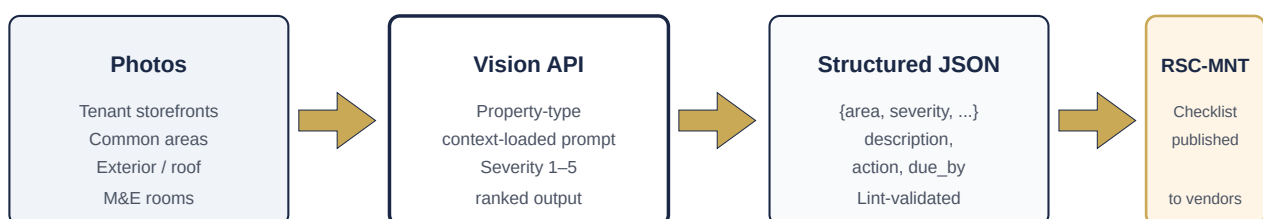
The AI defect-analysis layer

This is where the modern stack pulls ahead of the traditional one. Vision-language AI models in 2025 can read a property photo and tell you what is wrong with it. Not perfectly. Not exhaustively. But accurately enough at scale to change the economics of inspection.

Models an operator can use today

- **GPT-4V / GPT-4 Turbo with Vision** (OpenAI API; approximately \$0.01 per image). Pass any captured photo with the prompt: *"You are a CRE property inspector. List every visible deficiency in this image, ordered by severity. For each item, give (a) what you observe, (b) likely cause, (c) recommended action, (d) urgency on a 1 to 5 scale."* Output is structured text you can parse to JSON.
- **Claude 3.5 Sonnet Vision** (Anthropic API; comparable cost). Stronger on nuanced written analysis and on structured output via tool use. Use this when your prompt asks the model to return JSON directly — Claude is more reliable at strict JSON output than GPT.
- **GPT-4o with batch processing**. Same model as GPT-4 Turbo with Vision but supports batched submission for hundreds of captures per session. Use this when you have a full 200-photo property walk and want results in one API call.
- **Custom CV models** for specific defect classes — a Detectron2 model trained on roof damage, for example. Overkill for most operators. Cite as a future option if your centre has a recurring defect class you want to model on.

Workflow



Each photo in the property set runs through the same pipeline. Two hundred photos at \$0.01 each is roughly \$2 in API costs for full coverage of a 90,000 sqft centre. The operator's job is to lint the output, not generate it.

Each tenant photo, common-area photo and exterior photo goes through this. Two hundred photos at roughly \$0.01 each is approximately \$2 in API cost for full coverage of a single center. The output is structured JSON you can template into a checklist and push into your RSC-MNT (Maintenance Control Pack) and RSC-TRK (Takeover Tracker) tools.

The AI does what a 20-year inspector's eye does, in seconds, against your entire photo set. It does NOT tell you about hidden problems you didn't photograph, code compliance beyond what's visible, or anything requiring physical access. Use it for what it's good at — surfacing deficiency patterns at scale — and trust your physical walkthrough for the rest.

CHAPTER 06

From scan to video: building the property walkthrough

The original ask was a video of the property. There are three practical paths an operator can run in 2026 without hiring a videographer. Each is suited to a different use case.

Path A — Capture-in-one video walks

Modern phones can shoot 4K walkthroughs in long takes. The "Matterport-style" walkthrough comes from steady walking plus voiceover notes plus date-time stamps. The result is a property video that lives on your phone, costs nothing to produce, and looks professional if you shoot at golden hour and walk smoothly. Use this for daily maintenance documentation — the video becomes part of the property record.

Path B — Generated from a 3D model

- **Luma AI Genie** (approximately \$30/month). Generates cinematic walkthrough videos from photos or 3D models. The best current option for producing marketing videos and lender visuals from your photogrammetry output.
- **Runway Gen-3** (approximately \$15/month). Image-to-video. Useful for hero shots of completed work — punch-list items successfully resolved, transformation of a vacant unit, before/after comparisons for board meetings.
- **NVIDIA Instant NeRF** (free; requires a GPU). Converts photos into neural radiance fields you can navigate. Slower to process than photogrammetry but the output quality is genuinely cinematic. Worth setting up a dedicated GPU workstation if you plan to produce marketing walkthroughs quarterly.

Path C — 360° video

Ricoh Theta Z1 or Insta360 X4 produce 360° video that any phone, headset or browser can play. Useful for documenting a property at a specific point in time — "this is what the common-area looked like on 14 August 2026" — and for board members who want to navigate the property remotely before a meeting.

Recommendation: Path A for daily maintenance documentation. Path B for marketing and board presentations. Path C for archival snapshots you want to navigate remotely.

CHAPTER 07

The maintenance checklist factory

Turning the AI analysis output into an actionable checklist is a five-step pipeline. The operator's job at this stage is configuration, not work — the model produces the candidate list, the operator reviews every flagged item before dispatching a vendor.

1. **Run GPT-4V or Claude over all captured photos.** Upload them as a batch. One prompt per session with the property context baked in.
2. **Aggregate findings into a JSON structure.** Schema: `{area, severity, description, recommended_action, due_by}`. Validate the schema before pushing downstream.
3. **Template the structure into your Maintenance Control Pack.** The existing RSC-MNT deliverable is the receiving bucket — its inspection templates line up cleanly with the AI output format.
4. **Push to your Takeover Tracker Workbook (RSC-TRK)** for tasks with due dates assigned. This is the handoff from inspection to operations.
5. **Export to PDF for distribution to vendors and staff.** The PDF is also appended to your AI Drive archive so you have a date-stamped record of the inspection.

Example output: Magnolia Plaza (90,000 sqft, Acadiana)

Area	Severity	Description	Action	Due by
Anchor store roof, SE quadrant	4	Visible ponding water + staining on soffit	Roof inspection by certified roofer	14 days
Common-area bathroom, NE	3	Two fixtures leaking at handle	Plumbing dispatch	7 days
Parking lot, west row	2	Faded ADA-compliant striping	Re-stripe lot	30 days
Tenant #14 storefront	3	Damaged signage, broken fixture	Notify tenant; require repair under lease	21 days
HVAC roof unit #3	4	Visible corrosion + refrigerant staining	HVAC vendor diagnostic	14 days
Common-area hallway lighting, row B	2	Three fixtures flickering	Electrical dispatch	14 days
Rear delivery area	2	Pavement cracking, no pooling	Schedule sealcoat for next quarter	90 days
Tenant #07 storefront	1	Window decals slightly faded	Notify tenant; cosmetic, no urgency	60 days
Main entrance matting	3	Wear pattern exposing substrate	Replace matting	21 days
Sidewalk, building perimeter	2	Hairline cracking, no trip hazards	Monitor; surface seal next maintenance cycle	90 days

This is the output format. Whether you generate it once a year or once a quarter, you have a defensible record of property condition that survives personnel changes, vendor transitions and lease disputes. That is what the AI gives you that the traditional walkthrough cannot.

CHAPTER 08

Cost model: what this actually costs an operator

Honest cost breakdown for an operator with one shopping center. The numbers below are defensible against public pricing as of August 2026 — any operator can build a similar model for their own center.

Item	Cost	Notes
iPhone Pro (existing or upgrade)	\$0 or up to \$1,299	Most operators already have this. Upgrade justification: \$700–\$1,300 for one year's worth of avoided inspections.
Polycam Pro subscription	\$15 / month	Only needed during active capture cycles. Pause between quarterly inspections if you want.
OpenAI or Anthropic API	~\$10 / month typical	Variable by capture volume. A full property scan of ~200 photos runs about \$2 per scan in API costs.
Sketchfab Pro (optional)	\$15 / month	For private hosted models. Skip if you only need local files.
Luma AI Genie (optional)	\$30 / month	Only for video generation. Skip for a maintenance-only workflow.
Drone (DJI Mini 4 Pro)	\$759 one-time	Optional. Skip if you only need interior and ground-level exterior.
Grand total — first-year cost	~\$1,300 + ~\$60/month ongoing	The ongoing \$60/month covers Polycam, AI API spend and Sketchfab.

Compare against: a physical inspection by a CRE inspector runs roughly \$1,500 to \$4,000 per visit, before any remediation cost is dispatched. If your centre requires quarterly walkthroughs, that is \$6,000 to \$16,000 per year in inspector fees alone. The modern stack pays for itself in the first quarter of use and runs at a tenth the annual cost thereafter.

This is the strongest economic case in the operator's toolkit right now. The technology is mature. The cost is negligible compared to traditional alternatives. The output is institutional. The only thing stopping most operators from running it is the absence of a written workflow that ties the pieces together — which is the problem this article solves.

CHAPTER 09

Workflow: a 30-day starter for any shopping center

A practical rollout an operator can complete in a single calendar month without disrupting centre operations. Each week's output feeds the next week.

Week 1 — Setup

Install Polycam. Sign up for the OpenAI or Anthropic API key. Capture one full walkthrough of one vacant unit in your center. Process the photos and LiDAR data to a textured mesh. Export to a Sketchfab-hosted private model so you have a link your team can navigate.

Week 2 — Add the AI Layer

Take fifty captures from the same vacant unit and run them through GPT-4V or Claude with the property-type prompt from Chapter 05. Save the analysis JSON. Convert one of the analyses to a one-page PDF so you can show an owner, board member or vendor what the output looks like before scaling.

Week 3 — Capture Site-Wide

Walk the entire common area plus the exterior (and the roof if you have the drone). Capture two hundred-plus photos over one to two days. Process the captures to the same model. Run the AI analysis on the full photo set.

Week 4 — Generate the Property Condition Report

Bundle the AI analysis output into your RSC-MNT (Maintenance Control Pack). Distribute the candidate checklist to the operations lead and key vendors. Review and approve the action list. Save a dated snapshot in your AI Drive archive as `RSC-COND-2026-08-XX-v1.0.pdf` for the year-end record.

Once you've done it once, every subsequent capture builds on it. The model becomes the property's institutional memory. The 3D scan from Q3 2026 becomes the baseline against which Q4 captures are diff'd.

CHAPTER 10

Limitations and what NOT to trust the AI on

Honest section. This workflow is powerful but it is not magic. Knowing what it cannot do is as important as knowing what it can.

What the AI does NOT do

- Read structural drawings or as-builts beyond what a photo shows.
- Detect mold, electrical issues or plumbing problems that are inside walls.
- Verify code compliance (zoning, ADA, fire) — those need a human inspector or specialist.
- Replace a structural engineer, environmental consultant or licensed inspector.
- Always get it right. Every AI-flagged item should be verified by a human before a vendor is dispatched.

What the AI DOES do

- Surface patterns humans consistently miss when reviewing large photo sets.
- Order findings by severity consistently, the same way every quarter.
- Provide a written record of property condition at a specific moment in time — defensible against future disputes.
- Generate an initial checklist for vendor dispatch in minutes, not days.
- Scale — two hundred photos processed in roughly the time it takes to walk the property once.

The model is your institutional memory, not your conscience. It will confidently flag the wrong things sometimes. It will miss the right things sometimes. The operator's job is the review — never blind-dispatch an AI-only finding without a human looking at the actual photo first.

CHAPTER 11

Sample checklist output

The full maintenance checklist output for *Magnolia Plaza* — a 90,000 sqft open-air center in Acadiana used here for illustration. The structure below is the canonical template. Your property's name replaces Magnolia Plaza; the categories and severity distributions stay the same.

Header

MAGNOLIA PLAZA · Quarterly Property Condition Report

Inspection date: 14 Aug 2026 · Operator: Adam Abdalla · Inspector: AI-assisted (GPT-4V / Claude 3.5 Sonnet) · Total findings: 47 · Center size: 90,000 sqft · Vacancy rate: 6.7%

Summary by severity

Severity	Count	Severity	Count
5 — Immediate action	2	3 — Routine fix	14
4 — Schedule within 14 days	9	2 — Schedule within 30 days	13
5 — Schedule within 60 days	6	1 — Cosmetic / informational	3

Summary by category

Category	Count	Category	Count
Roof / eaves	5	Parking / pavement	7
HVAC / mechanical	6	Tenant storefronts	11
Plumbing (common areas)	3	Lighting / electrical	6
Signage	4	Landscape / exterior	5

Top ten findings (full table excerpt)

Below is the candidate list as it appears in the AI-assembled output, human-reviewed then templated into the RSC-MNT format. Every dispatched action has a corresponding entry in RSC-TRK for tracking.

#	Area	Sev	Description	Action
1	Anchor roof, SE quadrant	4	Ponding + soffit staining	Certified roofer inspect
2	HVAC roof unit #3	4	Corrosion + refrigerant stain	HVAC vendor diagnostic
3	Common-area bathroom, NE	3	Two leaking fixtures	Plumbing dispatch
4	Tenant #14 storefront	3	Damaged signage	Notify tenant
5	Main entrance matting	3	Substrate exposed	Replace matting
6	Hallway lighting, row B	2	Three fixtures flickering	Electrical dispatch
7	Parking lot, west row	2	Faded ADA striping	Re-stripe lot
8	Rear delivery pavement	2	Hairline cracking	Sealcoat next quarter
9	Sidewalk, perimeter	2	Hairline cracking	Monitor; seal next cycle
10	Tenant #07 storefront	1	Faded window decals	Notify tenant; cosmetic

The full 47-finding table is included in the operator's working RSC-MNT document. This page shows the format and the top-ten exemplar. Distribute the full list to the operations lead and key vendors; archive the candidate list as [RSC-COND-2026-08-14-v1.0.pdf](#).

CHAPTER 12

Appendix — full tool list with prices

The canonical operator's toolkit for the modern pocket-inspection workflow. Every tool listed below is something an operator can buy, subscribe to, or sign up for today. Pricing is approximate and rounded for clarity.

Tool	Category	What it does	Cost	Tier
Polycam	Photogrammetry + LiDAR	Mobile app — produces textured 3D models, supports LiDAR fusion on supported devices	Free / \$15 / mo Pro	Pro
Scaniverse	Photogrammetry	Mobile app — photogrammetry with Gaussian splatting support	Free (commercial separately)	Standard
KIRI Engine	Photogrammetry	Mobile app — interior-oriented walkthrough mode	Free / freemium	Standard
RealityCapture	Photogrammetry (desktop)	Survey-grade reconstruction from photos	Free (non-commercial) / ~\$3,500 perpetual	Advanced
Metashape	Photogrammetry (desktop)	Pro-grade reconstruction used by surveyors	~\$179 standard / ~\$3,499 pro	Advanced
Luma AI	Video generation	Generates cinematic walkthrough videos from photos or models	~\$30 / mo Genie	Optional
Runway Gen-3	Video generation	Image-to-video for hero shots and before/after	~\$15 / mo	Optional
Sketchfab	3D model hosting	Hosts and embeds GLB / splat models, shares a link	Free / ~\$15 / mo Pro	Standard
OpenAI API (GPT-4V / GPT-4o)	Vision AI	Analyzes property photos, returns structured deficiency reports	~\$0.01 / image	Pro
Anthropic API (Claude 3.5 Sonnet)	Vision AI	Analyzes property photos with strong JSON adherence	~\$0.01 / image	Pro
DJI Mini 4 Pro	Drone — visual	Visual photogrammetry; roof / exterior capture	~\$759 one-time	Optional
Skydio 2+	Drone — autonomous	Close-quarter autonomous flight for tight sites	~\$1,000–\$1,500	Optional
DJI Mavic 3 Enterprise + RTK	Drone — survey	Survey-grade accuracy for lender-grade aerial maps	~\$5,500	Advanced
Ricoh Theta Z1	360° camera	Best-in-class fast interior panoramas	~\$1,000	Optional
Insta360 X4	360° camera	Cheaper 360° alternative; good enough for archival	~\$400	Optional
Matterport Pro3	Dedicated LiDAR scanner	Industry-leading automated walkthrough tool	~\$5,500	Advanced

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